



Compact hybrid reluctance based tip/tilt actuator for large range fast steering mirrors

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ABSTRACT

This work presents a compact tip/tilt fast steering mirror (FSM) based on a linearized hybrid reluctance actuation topology. The actuation principle is analyzed using a magnetic equivalent circuit model and finite-element method simulations. The proposed design achieves a large range of $\pm 5/10$ deg (mechanical/optical) with an optical aperture of 1×1.5 inch, delivering higher torque and linearity compared to the state of the art. To maintain compactness, the system uses magnetically coupled axes while achieving a decoupling of over 30 dB. Independent control loops for each yield position bandwidths of 1.06 kHz and 1.13 kHz, enabling a precision below 1.2 mdeg, and Lissajous scanning up to 110 Hz with 4 deg. Overall, the compact FSM demonstrates the highest range-bandwidth product (+22%) with the largest range of hybrid reluctance actuated systems.

1. Introduction

Fast steering mirrors (FSMs) are a key component in high-performance optomechatronic systems where high-speed and precise beam steering is required. By dynamically adjusting the tip/tilt angle of a mirror, the reflected beam can be steered for pointing or scanning applications. Pointing applications require high precision for accurate beam steering and high bandwidths to compensate for high-frequency disturbances. Whereas, scanning applications require highly dynamic beam steering for large amplitudes. A large range can be beneficial in both applications due to the extended field of view. Often, fast scanning and precise pointing is performed by the same FSM, as in free-space optical communication [1,2], backscanning step and stare imaging [3], or a single FSM is designed and produced for various applications, such as light detection and ranging (Lidar) [4], adaptive optics [5], material processing [6], beam stabilization and telescope systems [7].

To meet the requirements for these applications, a high torque tip/tilt actuator with large range is needed to achieve high bandwidth, precision, and a large field of view. Furthermore, high efficiency minimizes thermal distortions of the mirror surface and ensures low power consumption. In general, piezoelectric actuators are employed for applications requiring high bandwidths and small ranges, whereas Lorentz actuators are used for larger ranges with moderate bandwidths. However, hybrid reluctance actuators achieve high bandwidths and a relatively high range, being a promising actuation technology for FSMs [8]. In addition, hybrid reluctance actuation obtains forces higher

than Lorentz actuators at equivalent volume, enabling the design of compact and highly integrated systems [9].

The drawback of hybrid reluctance actuation lies in the non-linearity regarding torque-to-current relationship and the stiffness, causing an operating-point dependent performance [10,11]. While some applications exploit this non-linear behavior for zero-power gravity compensation [12] or to adapt the stiffness for vibration isolation [13], a linear system response is typically advantageous in FSMs [14]. In non-linear systems, the transmission characteristics of external disturbances can vary with the operating point, potentially leading to a redistribution of the disturbance across the frequency spectrum. Especially for resonant scanning applications, the properties of an operating point-dependent system alter the mechanical resonance and reduce efficiency [15,16]. Moreover, compensating for such non-linearity through control design requires the use of non-linear feedback and/or feedforward compensators [17,18], which in turn requires accurate system identification and significantly increases control complexity.

Advanced air gap geometries can substantially reduce the non-linearity of hybrid reluctance actuators at the cost of reducing the torque-to-current ratio by a factor of three [19]. A novel actuator topology for hybrid reluctance actuators overcomes this limitation and obtains higher linearity and higher torque output at equivalent size compared to the conventional topology [20]. This topology has been verified only in one degree-of-freedom and a range of 1 deg, however,

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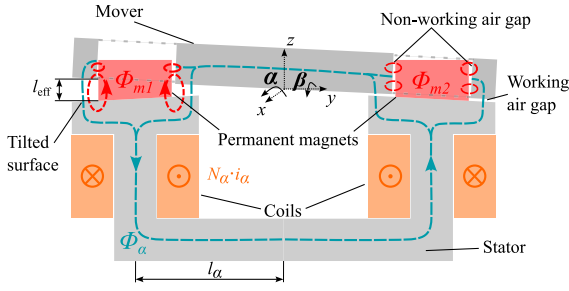


Fig. 1. Principle of the linearized hybrid reluctance actuator with variable effective length of the permanent magnets for the bias flux adaption. The permanent magnets plunge into the ferromagnetic mover, which partially short-circuits the upper part via the non-working air gap. The superposition of coil and bias flux results in a pull/pull force at each end of the mover.

many applications require beam steering in tip/tilt direction with larger range.

The contribution of this paper is a high-performance tip/tilt FSM design that achieves the largest angular range among hybrid reluctance actuated systems, while maintaining a compact form factor. The topology for improving linearity and the extension for tip/tilt motions is proposed in Section 2. The actuator is modeled by a magnetic equivalent circuit in Section 3 and evaluated by finite-element-method simulations in Section 4. A tip/tilt flexure is designed and an experimental setup with a prototype of the proposed design is built in Sections 5 and 6. Finally, the system is evaluated in Section 9, and the performance is compared to the state of the art in Section 10.

2. Linearized hybrid reluctance actuator principle

The principle of the linearized hybrid reluctance actuator for one degree-of-freedom [20] is shown in Fig. 1.

Hybrid reluctance actuation uses bias fluxes $\Phi_{m1,2}$, introduced by one or multiple permanent magnets, to enable bi-directional forces in a pull/pull configuration with the coil flux Φ_α and to linearize the torque-to-current ratio around the center position. In the conventional topology with one permanent magnet in the center, the inherent non-linearity arises from a non-linear increase in bias flux in the direction of deflection, e.g. a tilt in the α -direction leads to a disproportional increased bias flux in the left air gap [19].

The linearized topology has a permanent magnet at each working air gap to introduce the bias flux. The upper part of the permanent magnet is plunged into the ferromagnetic mover, and is partially short-circuited by the surrounding ferromagnetic material. This geometry provides a permanent magnet with a variable effective length l_{eff} , which leads to a linear adaptation of the bias flux in dependence of the deflection angle α . To minimize spatial saturation effects, the stator surface at the working air gaps are tilted by the maximum deflection angle. In the ideal case, this approach results in a linear torque-to-current ratio, as well as torque-to-deflection ratio. Experiments and finite-element method simulations revealed a minor non-linearity, caused by undesired reluctances. The linearized topology showed an enhanced linearity and higher torques at an equivalent actuator size, compared to the state of the art.

When designing a tip/tilt actuator based on a single degree-of-freedom actuator, several approaches can be considered to achieve the desired angular motion. A serial kinematic approach involves the serial connection of two single degree-of-freedom actuators. Although this allows for simple and fast system design, it introduces a greater inertia for the first actuator in the kinematic chain, yielding a significantly faster and slower purely based on this design approach. In contrast, a parallel kinematic configuration uses two actuators acting on one moving part, contributing equal inertia for tip and tilt, making parallel kinematics

Table 1

Targeted specifications of the FSM.

FSM specifications	
Optical aperture	38.1 x 25.4 mm ²
Angular range (mechanical/optical)	±5/10 deg
Bandwidth	> 1 kHz
Resolution	< 2 m deg

preferable in FSMs [21–23]. The magnetic circuit can be coupled to reach a highly integrated system design and compact size [24]. Since this work aims for a compact and high dynamic tip/tilt system, a parallel kinematic approach with a magnetically coupled system design is pursued. To enable rotation around the x- and y- two equivalent actuators are aligned orthogonally. Both actuators are mechanically and magnetically connected at the center of the mover, as illustrated in Fig. 3.

FSMs are commonly aligned at a 45 deg angle relative to the detector or laser source to enable compact optical systems. To obtain a clear aperture of 1 inch at a 45 deg angle, an elliptical mirror is required. The elliptical mirror introduces asymmetries of the axes, but enables the study of two different actuator lever lengths in terms of linearity in this design. Table 1 summarizes the targeted specifications, with particular emphasis on the large range for a hybrid reluctance actuated tip/tilt system, compared to the state of the art [8].

3. Magnetic equivalent circuit

To analyze the influence of coupling two actuators in regard to linearity of the actuator, a magnetic equivalent circuit model is created. Due to the symmetric design each, the magnetic equivalent circuit consists of two equivalent branches for each, resulting in a total of four branches with a stray reluctance in the center R_s , as shown in Fig. 2.

The torque-generating working air gaps, that are described by the reluctances

$$R_{1,2}(\alpha) = \frac{l_{\text{eff}1,2}}{\mu_0 A} \quad \text{with} \quad l_{\text{eff}1,2} = l_{\alpha 0} \mp l_\alpha \sin \alpha, \\ R_{3,4}(\beta) = \frac{l_{\text{eff}3,4}}{\mu_0 A} \quad l_{\text{eff}3,4} = l_{\beta 0} \mp l_\beta \sin \beta, \quad (1)$$

with the permeability of vacuum μ_0 , the cross-section of working air gaps A and the effective length of the air gap l_{eff} . The effective length is modeled by the initial air gap length l_0 ($\alpha = \beta = 0$ deg) and the corresponding lever length l of each. The bias flux in each working air gap is modeled by a magneto motive force of the permanent magnets with an inner reluctance, defined by

$$\Psi_{m_i}(\alpha, \beta) = H_c l_{\text{eff}i} \\ R_{m_i}(\alpha, \beta) = \frac{l_{\text{eff}i}}{\mu_0 A_m} \quad (2)$$

with the coercivity H_c , the effective length of each working air gap and the cross-section of the permanent magnets A_m . For torque-generation, the bias fluxes in opposing gaps are super-positioned with the coil fluxes. Therefore, the magneto motive forces of the actuation coils Ψ_c are defined by

$$\Psi_{c1,2}(i_\alpha) = \frac{N_\alpha i_\alpha}{2} \\ \Psi_{c3,4}(i_\beta) = \frac{N_\beta i_\beta}{2} \quad (3)$$

with the total number of turns $N_{\alpha,\beta}$ for each. To provide an equivalent coil flux in both air gaps of each, both coils of each are electrically connected in series.

By applying Gauss's and Ampere's law on the magnetic equivalent circuit in Fig. 2 with Eqs. (1)–(3), the magnetic fluxes in the working air gaps can be generally rewritten to functions of

$$\Phi_i(\alpha, \beta, i_\alpha, i_\beta) \quad (4)$$

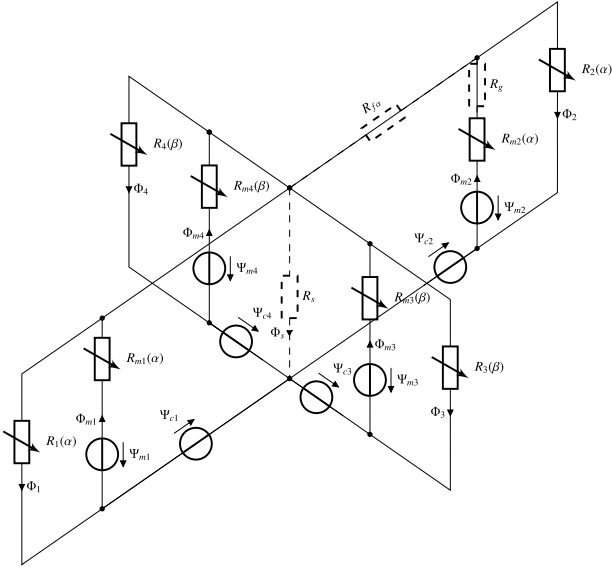


Fig. 2. Magnetic equivalent circuit of a magnetically coupled linearized HRA with the reluctances R , the magneto motive forces Ψ and the fluxes Φ with solid lines. The extended magnetic equivalent circuit is depicted only for branch $i = 2$ with the dashed reluctance.

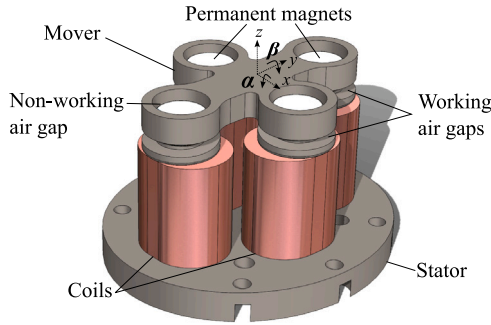


Fig. 3. CAD-model of the linearized tip/tilt hybrid reluctance actuator.

with the deflection of each α , β and the corresponding actuation currents $i_{\alpha,\beta}$. The remaining parameters of the magnetic equivalent circuit model are constant. Due to the magnetic coupling, the fluxes are depending on the actuation current and the deflection angle of the other. In order to model the torque output of the magnetically coupled tip/tilt actuator, the reluctance force at each working air gap is evaluated [16] and transformed by the leverage of the mover $l_{\alpha,\beta}$ into a torque for each, resulting in

$$\begin{aligned}\tau_{\alpha}(\alpha, \beta, i_{\alpha}, i_{\beta}) &= \frac{\Phi_1^2(\alpha, \beta, i_{\alpha}, i_{\beta}) - \Phi_2^2(\alpha, \beta, i_{\alpha}, i_{\beta})}{2\mu_0 A} l_{\alpha}, \\ \tau_{\beta}(\alpha, \beta, i_{\alpha}, i_{\beta}) &= \frac{\Phi_3^2(\alpha, \beta, i_{\alpha}, i_{\beta}) - \Phi_4^2(\alpha, \beta, i_{\alpha}, i_{\beta})}{2\mu_0 A} l_{\beta}.\end{aligned}\quad (5)$$

Due to the predominant orthogonality to the rotation axes of the fluxes Φ_{m_i} in the non-working air gaps, they are not considered in the torque-generation. So far, functional required components of the hybrid reluctance actuator are included in the model, shown by the solid lines in Fig. 2. To improve the model accuracy, parasitic reluctances extend the magnetic equivalent circuit, depicted in Fig. 2 for the branch $i = 2$ by the dashed reluctances. These parasitic reluctances are the non-working air gap reluctance R_g , the reluctance of the ferromagnetic core $R_{f,\alpha,\beta}$ and the stray reluctance R_s to model the stray flux.

Numerical evaluation of the torque $\tau_{\alpha,\beta}$ from the extended magnetic equivalent circuit with the geometrical parameters from the CAD-Model

and the material parameters adopted from subsequent finite-element-method simulations, are shown in for the α - Fig. 4(a) and the β - in Fig. 4(b) by the solid curves for currents in a range of ± 1 A, respectively. Due to the larger diameter of the α - and the resulting higher inertia, a larger leverage l_{α} is chosen. This provides higher torques for the α -. Also, a higher non-linearity than the β - is observable, especially towards the maximum deflection. To investigate cross-talk, the torque changes of the tip/tilt axes are numerically evaluated for actuation currents of 0..1 A in the respective other. This results in maximum tip/tilt torque changes of 5/3.7 %, indicating a minor cross-talk between the axes in the magnetic domain. Ideally, the torques of each are decoupled and can be linearly modeled by

$$\begin{aligned}\tau_{\alpha,\text{lin}}(\alpha, i_{\alpha}) &= \kappa_{\alpha}\alpha + k_{m\alpha}i_{\alpha} \\ \tau_{\beta,\text{lin}}(\beta, i_{\beta}) &= \kappa_{\beta}\beta + k_{m\beta}i_{\beta}\end{aligned}\quad (6)$$

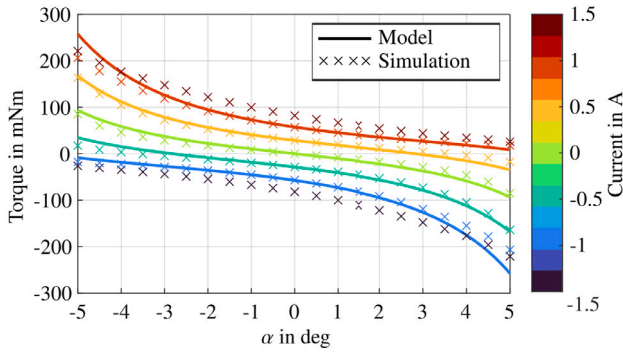
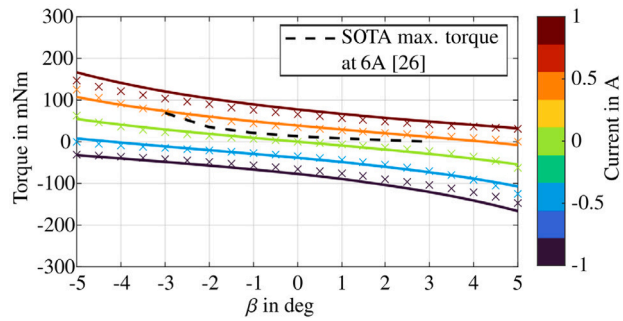
with the negative stiffness $\kappa_{\alpha,\beta}$, caused by the bias flux, and the torque-to-current ratio $k_{m\alpha,\beta}$ of each, respectively. For the one degree-of-freedom design, the magnetic equivalent circuit without parasitic reluctances is exact linear with a small angle approximation of $\sin(\alpha) = \alpha$ [25]. The proposed tip/tilt model with the identical simplifications shows analytically a non-linear behavior and is not decoupled. Therefore, the linearity of a magnetically decoupled tilt/tilt design is expected to be higher than a coupled version. However, for compact systems a magnetic coupling of the axes is expected due to the spatial proximity of the ferromagnetic parts and numerical evaluations of the model with the parameters from the subsequent system design revealed improved linearity, compared to the conventional hybrid reluctance structure with one permanent magnet in the center. A qualitative comparison to a state of the art (SOTA) system with the highest range of ± 3 deg with a diameter of 20 mm is visible in Fig. 4(b), showing a higher torque output, as well as a smaller stiffness change for the proposed magnetically coupled system. For reasons of compactness and the improvement in comparison to the SOTA, a magnetically coupled design is favored in this work.

4. Torque simulations

In order to verify the tip/tilt design and the proposed magnetic equivalent circuit, finite-element-method simulations (Maxwell 3D, Ansys Inc., USA) are conducted with the CAD-model, shown in Fig. 3. The simulations use non-linear B/H curves for the ferromagnetic components (VACOFLUX50, Vacuumschmelze) and the permanent magnets (NdFeB, Grade N45). Parametric analysis is performed for the torques $\tau_{\alpha,\beta}$ and the results are depicted by 'x' in Fig. 4.

The simulation results verify the magnetic equivalent circuit with the negative stiffness, caused by the bias flux, and increasing torques with positive actuation currents. Around the center region, the model shows small deviation, but towards larger deflections and high currents the model uncertainty increases, since saturation effects are not taken into account. Therefore, the torques are overestimated by the model for large deflections and currents, indicating partial saturation of the ferromagnetic parts in these operating regions. Conventionally, saturation effects are not beneficial regarding system linearity. However, in this case, they are used to limit the maximum torque towards large deflection.

By fitting the parameters of the linear torque model from Eq. (6), the stiffness κ , the torque-to-current ratio k_m and the root-mean-square (RMS) error for the fitted models of each are identified, shown in Table 2. Both axes show comparable torque-to-current ratios, but the tip is saturating at higher currents due to the larger initial air gap $l_{\alpha 0}$ and therefore achieves a higher maximum torque. The negative stiffness of the tip is higher due to the higher gradient of the working reluctance $R_{1,2}(\alpha)$, caused by the longer lever arm of the tip. The RMS error of the linear torque model for the tilt is significantly lower than that of the tip, caused by the shorter lever arm l_{β} and the smaller initial air

(a) Modeled and simulated torque of the tip ($\beta = i_\beta = 0$).**Fig. 4.** Torque simulation and model of the tip/tilt actuator at various operating points.**Table 2**

Simulation results for the stiffness and torque-to-current ratio of the linear torque model.

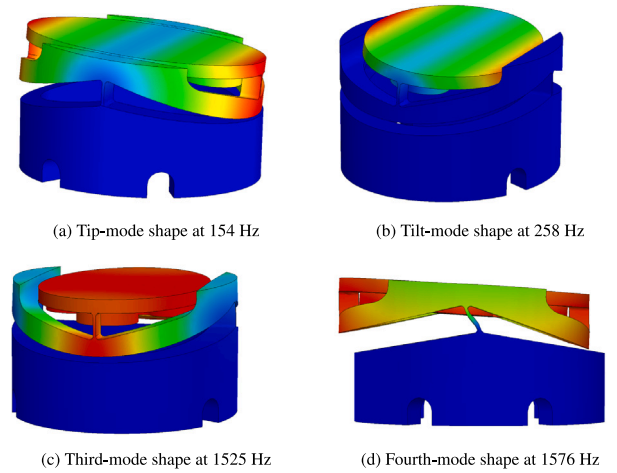
	k_m in mNm A^{-1}	κ in mNm/deg	RMS error in mNm
α	78.14	-14.67	16.77
β	78.29	-10.50	9.365

gap length $l_{\beta 0}$. Therefore, a higher linearity is observable for the tilt. In contrast, higher torque can be achieved by the tip, due to the longer lever arm $l_{\alpha 0}$. As consequence, shorter lever arms may be more suitable for highly linear systems at the cost of a reduced torque output.

5. Mechanical system design

The mirror requires a bearing that restricts the mover to tip/tilt motion. In addition, the actuator introduces negative stiffness in both axes, destabilizing the mover position. To stabilize the center position, a positive stiffness is required for the tip and tilt.

Flexure hinges are commonly used in FSMs to enable tip/tilt motion with stiffness and to restrict the remaining degrees-of-freedom by high stiffness [26–28]. To obtain the tip/tilt motion, two beam shaped hinges with fillets towards the rigid parts are serially connected, as shown in Fig. 5. A monolithic design, combining the housing, the flexure hinge and the polished surface, serving as mirror, ensures a highly integrated design and minimizes the number of mechanical components. The reflectivity of the mirror could be improved with a gold or silver coating in the infrared or visible range. However, for system identification and evaluation, the polished titanium alloy is sufficient. Ti-6Al4V is used for this component due to the relatively high yield strength to avoid damage of the flexure hinge at high deflection angles and a relatively low density for low inertia of the mover. Since manufacturing tolerances and static attraction force in vertical direction are present, the notch widths of the flexure are defined to 400 μm for both axes and the height/depth dimensions of the

**Fig. 5.** Modal analysis of the flexure hinge without the negative stiffness of the actuator.

notches are tuned to obtain the desired tip/tilt stiffness of $\kappa_{f\alpha\beta} > -\kappa_{\alpha\beta}$ for an open-loop stable center position. Higher-order mode shapes are considered to limit the position control bandwidth of FSMs [29]. Since the negative stiffness introduced by the permanent magnets reduces the effective stiffness in all degrees-of-freedom, the eigen-frequencies of the mode shapes are shifted towards lower frequencies. Therefore, the flexure hinge is tuned to shift the higher-order mode shapes above 1.5 kHz. The resulting dimensions (width $\times$ height $\times$ depth) of the hinges are $0.4 \times 4.5 \times 4 \text{ mm}^3$ / $0.4 \times 4.5 \times 3 \text{ mm}^3$ for the tip/tilt, respectively.

Static finite-element analysis is used to evaluate stiffness in the tip/tilt direction. The stiffness is required to be high to stabilize the center position and to obtain high stiffness in the non-actuated degrees of freedom, whereas it needs to be as low as possible for minimum power consumption for static deflection. This trade-off leads to stiffness of $k_{f\alpha} = 25.8 \text{ mNm/deg}$ and $k_{f\beta} = 19.8 \text{ mNm/deg}$. At maximum deflection the von Mises stress is evaluated to be 419/416 MPa for the tip/tilt axes.

Modal analysis shows, the first two mode shapes being defined by the tip/tilt stiffness and inertia of the moving part, providing the desired tip/tilt motion, shown in Fig. 5. The third mode shape is a trampoline mode, which is not observable motion for the tip/tilt position sensor. The fourth mode shape is a combination of a rotation around the tip and an in-plane motion. By reducing the height, the in-plane stiffness could be improved, but the mechanical stress is also increased, running into the boundary of the yield strength. This trade-off resulted in a fourth mode shape at 1.57 kHz. An equivalent fifth mode is observable for the tilt at 1.63 kHz. Both modes are expected to be observable due to rotational motion.

6. Experimental setup

For the experimental evaluation, a FSM prototype is implemented and integrated into a setup comprising a collimated laser, an optical position sensor based on an autocollimator principle [30,31], and power electronics supplied with $\pm 30 \text{ V}$ for driving the coil currents, as shown in Fig. 6. To compensate for the dynamics of the electrical subsystem, an analog PI controller with a -3-dB bandwidth of 15 kHz is used. The position sensor based on an autocollimator is used to sense the deflection angle of the laser beam. It consists of 635 nm collimated laser with 0.9 mW, a beamsplitter to enable the orthogonal alignment, a focusing lens and position sensitive device (Hamamatsu, S5991). The laser beam is partially reflected by the beamsplitter towards the FSM, which deflects the reflected laser beam. The deflected laser beam is partially transmitted through the beamsplitter and subsequently focused

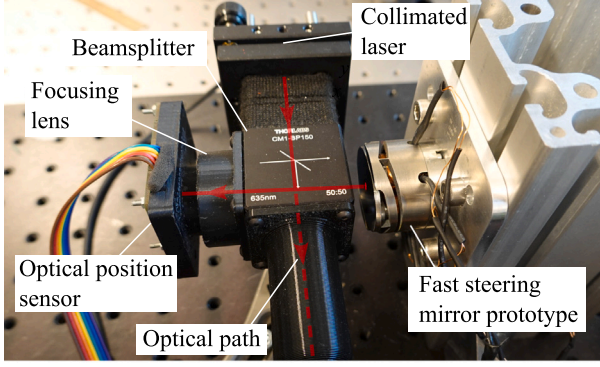


Fig. 6. The experimental setup includes the FSM prototype, which deflects the laser beam, resulting in a shift on the optical position sensor.

onto the optical detector plane by the focusing lens with a focal length $l_f = 25.4$ mm. The position on the detector x_{PSD} and y_{PSD} are used to determine the tip and tilt angle by

$$\alpha = \frac{1}{2} \arctan \frac{x_{\text{PSD}}}{l_f}$$

$$\beta = \frac{1}{2} \arctan \frac{y_{\text{PSD}}}{l_f} . \quad (7)$$

To compensate sensing errors due to misalignment, the autocollimator is calibrated in a 0.5 deg grid with a manual tip/tilt gimbal mirror (Thorlabs, GMB1). For the current measurements, an active current probe (Agilent, 1147 A) is used. A dSPACE MicroLab Box, operating at a sampling frequency of 100 kHz, is used as a rapid prototyping system for A/D conversion, real-time processing of position sensing data and the implementation subsequently designed position controller.

7. System identification in offset positions

With the experimental identification, the expected system behavior from simulation and model results are evaluated. To identify the non-linear behavior, the dynamic system response is measured at various offset positions from the current $i_{\alpha,\beta}$ to the corresponding deflection angle α, β . The identified frequency responses of the tip and tilt are shown in Fig. 7.

Both axes reveal a second order behavior with a spring-line at low frequencies, a resonance and a mass-line at above the resonance, which can be modeled for a general rotation γ by

$$P_{\text{sim}}(s) = \frac{\gamma(s)}{i_\gamma(s)} = \frac{k_{m\gamma}}{J_\gamma s^2 + D_\gamma s + \kappa_\gamma + \kappa_{f\gamma}} e^{-sT_d} \quad (8)$$

with the torque to current ratio k_m , the inertia J , the damping coefficient D , the total stiffness of $\kappa + \kappa_f$ and the time delay of the digital position processing T_d . The simulated system dynamics, with the parameters from the torque simulations in Table 2, the stiffness and inertia from the mechanical simulations, comply with the measurement results at the center position, verifying the torque simulations, as well as, the proposed magnetic equivalent circuit. Due to changing stiffness κ and torque to current ratio k_m of the tip/tilt actuator, major shifts of the spring-line (<22 dB), as well as, minor shifts of the mass-line (<3 dB) are observable. The position-dependent stiffness and torque-to-current ratio demonstrate the non-linearity of the system. The stiffness is reduced towards larger offset positions, which verifies the increasing negative stiffness of the actuator κ , as visible in Fig. 4. Both axes are open-loop stable for the identified range of ± 4 deg, which is in agreement with stiffness changes of the torque simulations.

An increased phase-loss φ_{core} is observable due to ferromagnetic core-losses, such as hysteresis and eddy current losses [25]. These inherent effects of ferromagnetic material based systems introduces

Table 3
Position control parameters.

	k_p	ω_I	ω_D	ω_t	ω_{DD}	ω_{tt}
α	5.24×10^{-1}	6.28×10^2	1.05×10^3	9.42×10^3	1.05×10^3	9.42×10^3
β	1.65×10^{-1}	1.26×10^3	1.26×10^3	1.13×10^4	1.26×10^3	1.13×10^4

another source of non-linearity, especially hysteresis effects can reduce the position control accuracy, which can be compensated by flux control [32]. This work uses a low coercivity material (100 A m^{-1}) with high electrical resistivity ($0.42 \mu\Omega \text{ m}$) to reduce the hysteresis and eddy current effects with respect to linearity, as well as reducing power dissipation. Asymmetries for the positive and negative deflection are observable on the spring-line due to mechanical and magnetic material tolerances of the prototype. Since electromagnetic actuated FSMs are conventionally controlled on the mass-line, the performance of the closed-loop controlled system is mainly influenced by the minor shifts of the mass-line. The decreasing stiffness for large deflection reduces the power consumption for static deflection.

To identify the cross-talk between the axes P_{ct} , the frequency response from the current $i_{\alpha,\beta}$ to the deflection of the other is measured in open-loop in the center position with the results, shown in Fig. 7. The decoupling at low frequencies is identified to be 37/30 dB for the tip/tilt, respectively. Significant decoupling of the axes is provided up to a frequency of 1.3 kHz, as visible at the significantly lower magnitude of the cross-talk P_{ct} compared to the main axes in Fig. 7, which is limited by the higher order modes at 1.3 kHz and 1.4 kHz.

8. Position control design

In order to obtain a FSM system, which follows an arbitrary reference trajectory, a feedback controller is designed for each. Since the decoupling of the axes is provided up to the targeted position control bandwidth of 1 kHz and the variations on the mass-line are small, a single-input-single-output (SISO) time-invariant position control is used for both axes.

To compensate the core phase losses φ_{core} , a proportional–integral–derivative (PID) controller with a second order derivative term (DD), defined by

$$C_{\text{pos}}(s) = k_p \left[\frac{\omega_I}{s} + \frac{1 + \frac{s}{\omega_D}}{1 + \frac{s}{\omega_t}} + \left(\frac{1 + \frac{s}{\omega_{DD}}}{1 + \frac{s}{\omega_{tt}}} \right)^2 \right] \quad (9)$$

is used to control the system [33,34]. The integral term compensates for static errors, the proportional term is tuned to obtain the desired cross-over frequency, and the derivative and second order derivative term is used to obtain sufficient phase lead in the cross-over region. According to the loop-shaping approach in [35], the control parameters are manually tuned to achieve a gain margin of 8 dB and phase margin of 50 deg for each axis. The parameters of the control design are summarized in Table 3. Since the gain variations identified in Fig. 7 are below 3 dB, with minor phase variations, the chosen margins are considered sufficient to compensate for undetermined variations in the control system.

For attenuation of the resonances, caused by the fourth, and fifth mode of the mechanical parts, notch filters [36]

$$N(s) = \frac{s^2 + 2D_N \zeta_N \omega_N s + \omega_N^2}{s^2 + 2\zeta_N \omega_N s + \omega_N^2} \quad (10)$$

are used. For the tip the notch frequency is $f_N = 1.3$ kHz, to attenuate the mode shape, in Fig. 5(d). The notch frequency for the tilt is defined with $f_N = 1.4$ kHz due to an equivalent mode shape. The open-loop transfer is defined by

$$L(s) = P(s)C_{\text{pos}}(s)N(s) \quad (11)$$

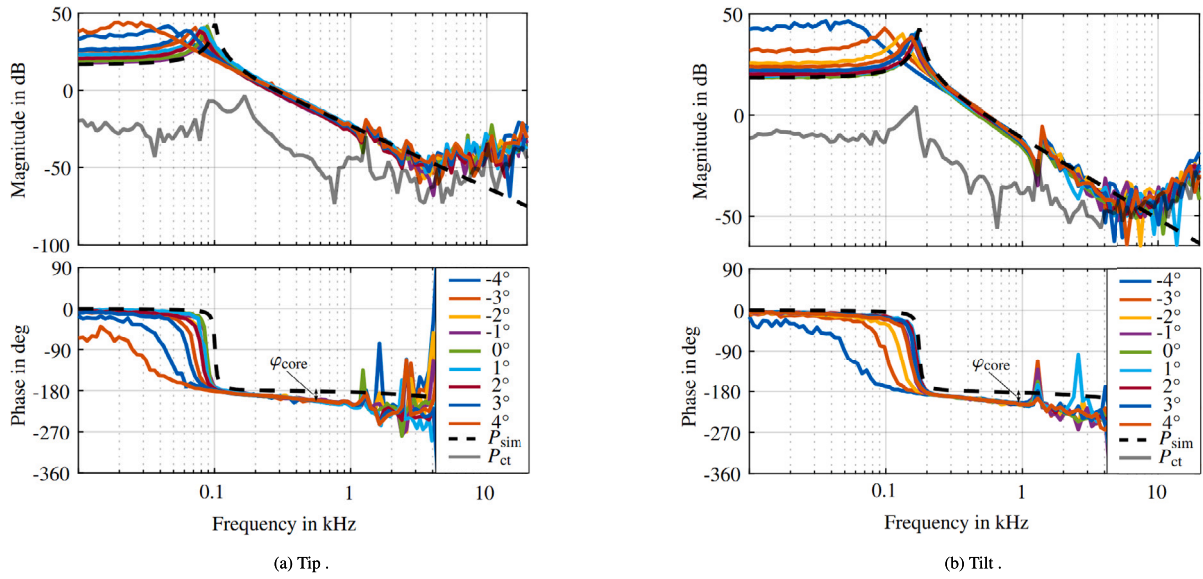


Fig. 7. Frequency response of the FSM prototype for the tip/tilt at various offset positions. The dynamic model P_{sim} , verifies the torque and mechanical simulations and the cross-talk P_{ct} shows significant decoupling of both axes.

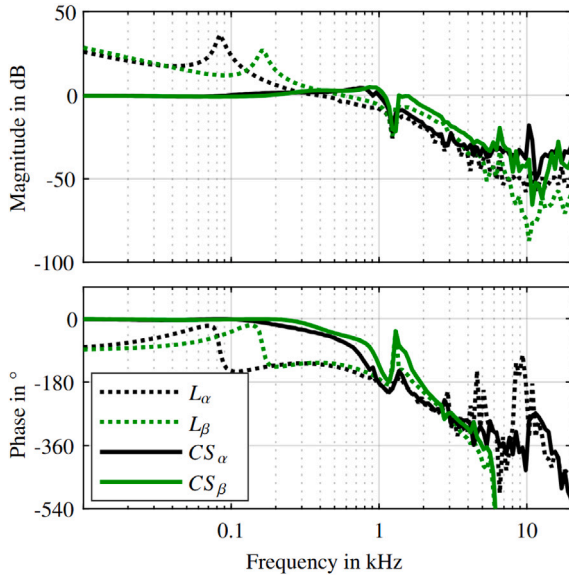


Fig. 8. Open-loop transfer functions $L_{\alpha, \beta}$ and the resulting complementary sensitivity functions $CS_{\alpha, \beta}$.

the plant $P(s)$, the proposed position control $C_{pos}(s)$, and the notch filter at the previously defined frequencies. The open-loop transfer functions $L_{\alpha, \beta}(s)$ at zero deflection offset are shown in Fig. 8.

The resulting control performance is demonstrated by the measurement of the complementary sensitivity functions $CS_{\alpha, \beta}(s)$, shown in Fig. 8. The tip/tilt axes exceed the targeted -3 dB position control bandwidth with 1.06 and 1.13 kHz, respectively. Both bandwidths are limited by the anti-resonance of the system at about 1.2 kHz, causing a reduced loop-gain and therefore the significant drop in the complementary sensitivity function. Due to the higher loop-gain of the tilt above the anti-resonance, reference tracking up to 1.7 kHz is achieved.

9. Experimental evaluation

Based on the position-controlled system, the performance is experimentally evaluated and subsequently compared to the state of the

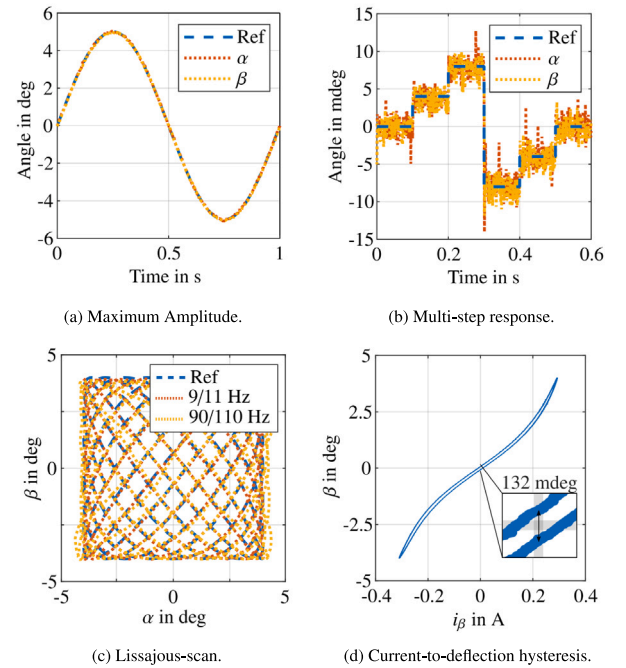


Fig. 9. Experiments with (a) the maximum amplitude of 5 deg, (b) a multi-step response with 4 mdeg steps and a 16 mdeg backstep, (c) Lissajous scans for two frequency pairs with a scan period of 1 and 0.1 s, and (d) the corresponding current-to-deflection hysteresis, which is compensated by the position control.

art. Fig. 9 demonstrates the maximum amplitude of ± 5 deg with a low-frequent sinusoidal reference trajectory, verifying the large range capabilities of the prototype in both axes. Both position controllers are stable during the experiment due to the sufficient phase and gain margins.

To evaluate the precision of the FSM system, a multistep response with 4 consecutive steps of 4 mdeg and a backstep of 16 mdeg are applied as reference position for both axes and the system response is shown in Fig. 9. The 4 mdeg steps are clearly distinguishable in

Table 4
Performance overview of hybrid reluctance actuated FSMs.

FSM specifications	Csencsics et al. [37]	Kluk et al. [38]	This work
Optical aperture in mm ²	20 × 20	20 × 20	38.1 × 25.4
Range (mechanical) in deg	±3	±0.2	±5
Bandwidth in kHz	1.5	10	1.1
Precision in mdeg	2	0.05	1.1
Volume in cm ³	24.1	169	29.1
Performance metrics			
Range-Bandwidth (RB) product in degkHz	4.5	2	5.5
RB-Aperture product in degkHzmm ²	1413	800	4197
Aperture-to-Volume ratio in m ⁻¹	13	2.4	26
Precision-to-Range ratio in ppm	333	125	110

the measurement results with the tip obtaining slightly higher peak errors. This asymmetry may be caused by the higher P-gain of the position control, which is required to obtain equivalent bandwidths, and an equivalent RMS error in both axes of the optical position sensor (0.65 mdeg). Whereas, by evaluating the RMS error in a static case at 0 deg, the tip/tilt axes perform quasi-equivalently with an RMS error of 1.2/1.1 mdeg, respectively, verifying the targeted system precision of < 2 mdeg.

Since scanning applications require high amplitudes at high frequencies, a Lissajous scan trajectory with amplitudes of 4 deg and scanning frequencies of 9/11 Hz and 90/110 Hz for the tip/tilt axes are evaluated. Fig. 9 shows the corresponding measurement results of one trajectory period (1 s/0.1 s). For the slower scan trajectory the reference can be followed with minor control errors, whereas for the faster scan higher errors are observable in the more inert tip. This is caused by the maximum output voltage of the power electronics of 30 V. Nevertheless, the experiments verify that the FSM is capable of large scans of up to at least 90/110 Hz. A full range sinusoidal scan with maximum currents of 1.5/1 A for the tip/tilt would be possible up to frequencies of 170 Hz/260 Hz based on the system identification in Fig. 7.

To analyze the influence of the hysteresis effects in the ferromagnetic material, the measured current-to-deflection response of the Lissajous-scan with 4 deg and 11 Hz is shown in Fig. 9(d). A clear hysteresis loop is observed, resulting in an offset of 132 mdeg at the origin, corresponding to a relative deviation of 1.32 % of the full range. In open-loop operation, this offset directly affects the mirror position. However, in closed-loop operation the position controller compensates for the hysteresis. Furthermore, the decreasing stiffness of the FSM with increasing deflection angle, identified in Fig. 4 and Fig. 7 causes the steeper system response at large deflections. Consequently, a current of only 0.3 A is required to achieve a deflection of 4 deg.

10. Performance comparison

In order to interpret the experimentally evaluated performance of the proposed system, a performance overview of hybrid reluctance actuated FSMs is shown in Table 4. The table comprises the specifications of systems with the largest range and the highest bandwidth from literature.

The proposed system in this work shows the largest optical aperture, as well as, the largest range, whereas the mirror from Kluk et al. [38], achieves the highest position control bandwidth and precision and Csencsics et al. [37] obtains the smallest volume of all designs.

Based on these specifications, several performance metrics can be derived. The range bandwidth product should represent the trade-off between the position control bandwidth and the range of FSMs. A comparison of 39 Lorentz, Piezo and hybrid reluctance actuated FSMs shows, that the mirror from [37] has the highest range-bandwidth-product [8]. This work improves this performance metric by 22%. An extension of this metric is the range bandwidth aperture product, used in [39], also considering the optical aperture and related inertia. With

the large aperture of this work, this performance metric is improved by a factor of 3. As a metric for compactness, the Aperture-to-Volume ratio is evaluated, since a larger optical aperture conventionally requires a larger volume. The volume of this work is slightly larger, compared to the compact design of [37], but due to the significantly larger optical aperture, compactness also improves by a factor of 2. Lastly, the precision normalized to the range is evaluated. As [38] and this work uses an optical position sensing method, the precision of both is comparable, whereas [37] obtained a lower performance.

In summary, the proposed compact hybrid reluctance actuated tip/tilt FSM enhances linearity and torque output, provides a large range of ±5/10 deg (mechanical/optical) and achieves a 22% improvement in the range-bandwidth product.

11. Conclusion

In conclusion, a novel FSM is designed based on a linearized topology for hybrid reluctance actuation. The working principle of the magnetically coupled axes is verified by the magnetic equivalent circuit model and finite-element-method simulations. The tilt with the shorter lever arm is preferable for higher linearity, whereas the tip with the longer lever arm achieves higher torque. The simulation results compared to the literature reveal higher torque and linearity, which are verified by the identification of the prototype system. A significant decoupling of > 30 dB between the axes is demonstrated, enabling independent position control of each, obtaining bandwidths of 1.06 and 1.13 kHz. Experiments demonstrate the large range of ±5/10 deg (mechanical/optical), a precision of 1.2/1.1 mdeg and 4 deg Lissajous scans up to 110 Hz. In comparison to the literature, this work achieves the highest range bandwidth product (+22%) with the largest range and optical aperture reported for hybrid reluctance actuated systems and maintains a compact size due to the magnetically coupled design. The full range bandwidth is currently limited by the output voltage of the power electronics of ±30 V, being part of future research to experimentally evaluate the limits of the FSM.

CRediT authorship contribution statement

Alexander Pechhacker: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tobias Schopf:** Writing – review & editing, Resources, Funding acquisition. **Ernst Csencsics:** Writing – review & editing, Supervision, Project administration. **Georg Schitter:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Alexander Pechhacker reports financial support was provided by Micro-Epsilon Messtechnik GmbH and Co. KG. Alexander Pechhacker has

patent Reluktanz-Aktor, Fast Steering Mirror und Verfahren pending to MICRO-EPSILON-MESSTECHNIK GmbH & Co. KG. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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